Ethan Johnston

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EDUCATION

Illinois State University, College of Arts and Sciences **Expected December 2026**
Master of Science in Quantitative Economics Cumulative GPA: 3.87/4.0
Relevant Coursework: Real Analysis, Advanced Linear Algebra, Microeconometrics, Advanced Macroeconomic Theory

Stephen F. Austin State University, Rusche College of Business **May 2025**
Bachelor of Business Administration, Finance; Minors in Economics and Mathematics Cumulative GPA: 3.967/4.0
Relevant Coursework: Calculus I-III, Ordinary Differential Equations, Linear Algebra, Speculative Markets

EXPERIENCE

Graduate Research Assistant, Illinois State University **August 2025 – December 2025**
- Built a reusable Python data pipeline automating API ingestion, record matching, and data validation workflows across 2,922 publication records and 215 authors
- Structured and validated a research dataset covering 606 individuals from U.S. Economics PhD programs, improving downstream analysis and reproducibility

Financial and Tech Sector Analyst, Student Investment Fund, Stephen F. Austin State University **August 2024 – May 2025**
- Developed a Python-based valuation support framework integrating Excel-driven financial models and deployed it as an interactive Streamlit application used to support portfolio allocation decisions
- Performed financial modeling and valuation analysis in Excel using regression-based factor models and statistical testing to evaluate positions within a $900,000 portfolio
- Presented quantitative analyses and investment recommendations to peer analysts, translating technical model output into actionable portfolio insights

QUANTITATIVE DEVELOPMENT PROJECTS

Synthetic Derivatives & Risk Engine for Sports Betting Markets **January 2026 – Present**
- Engineered a Python-based pricing and backtesting framework for 25 derivative instruments, building the full workflow from data ingestion through model generation across a 2.5M-observation dataset
- Built automated web-scraping and preprocessing pipelines across 600+ sources, standardizing large-scale market data for downstream quantitative analysis
- Developed a C# / .NET quantitative risk engine consuming Python-generated pricing outputs to perform Monte Carlo simulation, VaR/CVaR analysis, regime stress testing, and portfolio risk monitoring
- Improved simulated risk-adjusted returns by 29% through portfolio optimization using mean-variance and CVaR methodologies across 5,039 rolling evaluation windows

Analyzing High-Volume Trades in Dow Jones ETF **August 2024 – May 2025**
- Developed R-based data-processing utilities for chunking and analyzing intraday datasets exceeding 300,000 observations
- Modeled intraday price and return dynamics in Python using logistic regression and comparative distribution analysis across varying market conditions

RESEARCH PAPERS

Displacement and Task Composition (PSID/CPS) **February 2026 – Present**
- Conducting large-scale labor market analysis using PSID and CPS panel datasets (60M+ observations) with panel-data and causal inference methodologies

Service Quality and Digital Piracy: Evidence from Denuvo DRM **January 2025 – Present**
- Analyzing the relationship between digital rights enforcement and consumer demand using large-scale panel datasets and econometric modeling techniques

AWARDS & PRESENTATIONS

- Merit Fellow for Economics Department, ISU (2025–2026); Scott Elliot Graduate Scholarship, ISU (2025–26); Portfolio Management Award, SFASU (2024–2025); Highest Ranking Finance Senior, SFASU (2023–24); MEA 2026 and IEA 2025 Presenter

TECHNICAL SKILLS

Programming & Analytics: Python (pandas, NumPy, SciPy), R, Stata, MATLAB, JavaScript, SQL (basic), LaTeX, Git/GitHub
Financial & Data Engineering: Financial modeling, Excel workflow automation, ETL/data pipelines, API integration (REST/JSON), web scraping, large-scale data processing
Tools: Excel (pivot tables, advanced formulas, Data Analysis ToolPak), Word, PowerPoint, Streamlit